



NATIONAL UNIVERSITY OF MODERN LANGUAGES
ISLAMABAD

FACULTY OF ENGINEERING & COMPUTING
DEPARTMENT OF COMPUTER SCIENCE

END-TERM EXAMINATION – Fall 2025 – BSCS VII A&B (MORNING)

Roll No. CS-193

Subject: Computer Graphics
Allowed Time: 03 Hours

Instructor: Sabina
Total Marks: 50

Instructions: -

- Attempt all questions and return the question paper with the answer sheet.
- Clearly mention the question number before attempting it

QUESTION 1 [CLO-2, C3, PLO-3]

Part 1:

[5x2=10 Marks]

- You are developing a 2D graphics application where a map of a city is displayed on the screen. The application defines a window in world coordinates representing the area of interest: $(X_{wmin}, X_{wmax}, Y_{wmin}, Y_{wmax}) = (10, 70, 30, 90)$. The map is displayed on a viewport on the user's screen: $(X_{vmin}, X_{vmax}, Y_{vmin}, Y_{vmax}) = (50, 100, 20, 50)$. A point of interest in the city has world coordinates $(12.5, 45.0)$. Apply the corresponding viewport coordinates (S_x, S_y) for this point using the window-to-viewport transformation.
- Apply transformations on a triangle that has vertices at $A(2,3)$, $B(5,7)$, $C(6,2)$.
 - Scale the triangle by 2 along x-axis and 1.5 along y-axis.
 - Translate the scaled triangle by $(3, 4)$.
 - Draw a diagram showing the original and transformed triangle.

Part 2:

[10x2=20 Marks]

- You are developing a 2D graphics application that displays a portion of a map. A user draws a line segment from $R(5,4)$ to $S(10,8)$. However, the application only shows content within a rectangular clipping window defined by the vertices: $A(4,3)$, $B(9,3)$, $C(9,7)$, $D(4,7)$. To ensure only the visible portion of the line is displayed, the application must clip the line against the window boundaries. Apply the Cohen-Sutherland line clipping algorithm to determine the final visible segment of the line.
- A 2D graphics application draws a line from $P(0,6)$ to $Q(4,0)$ on a raster display. The application uses Bresenham's line algorithm to determine which pixels to light up. Rasterize the line and list all pixel positions step by step. Solve the steps for the algorithm to decide the next pixel at each step using the decision parameter. Mention why Bresenham is efficient for integer-based calculations as compared to DDA.

QUESTION 2 [CLO-3, C3, PLO-4]

Part 1:

[10x2=20 Marks]

- You have drawn a closed polygon on the screen. The teacher instructs you to fill it. To explain the required behavior, the teacher says: "Imagine a single pen starting from the bottom-left, moving horizontally to the right, marking only the pixels that lie inside the polygon. After finishing one horizontal line, the pen moves up to the next line and repeats the process until the polygon is fully filled." Construct the pseudocode to identify the polygon-filling algorithm that matches this behavior.
- You are creating a scene where the light source is very far away, such as the sun. The rays reaching the scene are essentially parallel. Organize and identify C++ OpenGL function used to define this light and apply material properties on it.
